

Wall Following Robot Project

README:

Project Overview

I implement a **robust wall-following mobile robot** using a PID controller in **CoppeliaSim** with the **Pioneer P3-DX** robot. The robot is capable of smoothly following **left or right walls**, handling **corners**, and maintaining a stable distance from obstacles. The focus of the project is on control stability, and perfectly follow the wall without any wall crashed.

Objectives:

- Implement a reliable wall-following behavior.
- Maintain a fixed distance from the wall.
- Handle sharp corners and narrow corridors.
- Minimize oscillations using PID control.

Tools & Technologies:

- CoppeliaSim (Edu Version)
- Pioneer P3-DX mobile robot
- Lua scripting language (As I also create a Txt file code which directly open on VS Code)
- Ultrasonic proximity sensors
- PID Controller (K_p , K_i , K_d)

Features:

- Left and right wall following support
- Smooth velocity and steering control
- Corner detection and correction